

# 14-subgroup: 子群 (Subgroup)

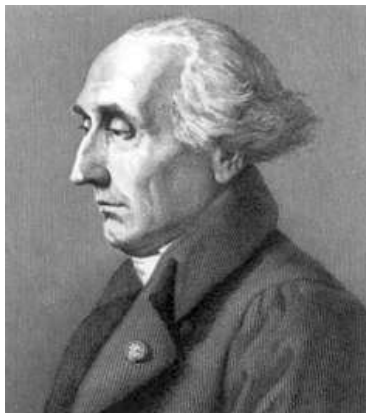
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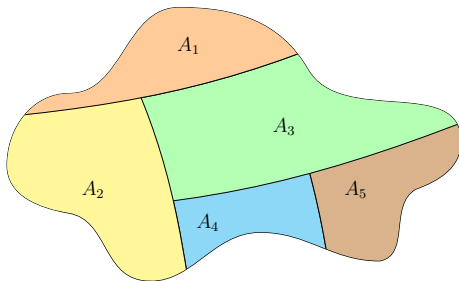


## Lagrange's Theorem



Joseph-Louis Lagrange (1736 ~ 1813)

## Lagrange's Theorem



Help us understand the structure of a group via its subgroups

### Definition (Coset (陪集))

Suppose that  $H \leq G$ . For  $a \in G$ ,

$$aH = \{ah \mid h \in H\}, \quad Ha = \{ha \mid h \in H\},$$

is called the **left coset** (左陪集) and **right coset** of  $H$  in  $G$ , respectively.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H \leq S_3$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)A_3 = (1\ 2\ 3)A_3 = (1\ 3\ 2)A_3 = A_3 \leq S_3$$

$$(1\ 2)A_3 = (1\ 3)A_3 = (2\ 3)A_3 = \{(1\ 2), (1\ 3), (2\ 3)\}$$

## Theorem

Suppose that  $H \leq G$ ,  $a, b \in G$ .

(1)

$$a \in aH$$

(2)

$$aH = H \iff a \in H$$

(3)

$$aH \leq G \iff a \in H$$

(4)

$$aH = bH \iff a^{-1}b \in H$$

(5)

$$\forall a, b \in G (aH = bH) \vee (aH \cap bH = \emptyset)$$

(6)

$$|aH| = |H| = |bH|$$

$$aH = H \iff a \in H$$

$$aH = H \implies a \in H$$

$$a \in H \implies aH = H$$

$$a \in aH = H$$

$$aH \subseteq HH \triangleq \{hh' \mid h, h' \in H\} = H$$

$$H = (aa^{-1})H = a(a^{-1}H) \subseteq aH$$

$$aH \leq G \iff a \in H$$

$$a \in H \implies aH \leq G$$

$$aH \leq G \implies a \in H$$

$$a \in H \implies aH = H \leq G$$

$$a \in aH \implies a^{-1} \in aH \implies e \in aH$$

$$\exists h \in H \ e = ah \implies a = h^{-1} \in H$$

$$aH = bH \iff a^{-1}b \in H$$

$$a^{-1}b \in H \iff a^{-1}bH = H$$

$$aH = bH \implies a^{-1}aH = a^{-1}bH \implies a^{-1}bH = H$$

$$a^{-1}bH = H \implies a(a^{-1}bH) = aH \implies bH = aH$$

$$\boxed{\forall a, b \in G \ (aH = bH) \vee (aH \cap bH = \emptyset)}$$

$$\forall a, b \in G \ (aH \cap bH \neq \emptyset \rightarrow aH = bH)$$

Take any  $g \in aH \cap bH$ .

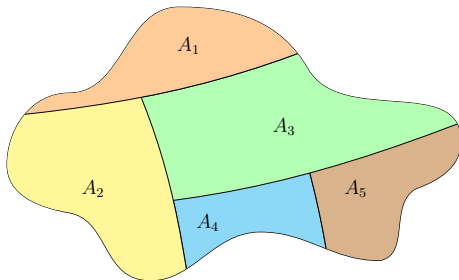
$$\exists h_1, h_2 \in H \ (ah_1 = g = bh_2)$$

$$aH = a(h_1H) = (ah_1)H = (bh_2)H = b(h_2H) = bH$$

$$|aH| = |H| = |bH|$$

$$\sigma : aH \rightarrow H, \quad ah \mapsto h$$

A balanced partition of  $G$  by its subgraph  $H$



$$\forall a, b \in G \ (aH = bH) \vee (aH \cap bH = \emptyset)$$

$$|aH| = |H| = |bH|$$

Definition (Index ( $H$  在  $G$  中的指标/指数))

$$G/H = \{gH \mid g \in G\}$$

$$[G : H] \triangleq |G/H|$$

Theorem (Lagrange's Theorem)

*Suppose that  $H \leq G$ . Then*

$$|G| = [G : H] \cdot |H|$$

## Theorem

$$H \leq G \implies |H| \mid |G|$$

There are *no* subgroups of order 5, 7, or 8 of a group of order 12.

## Theorem

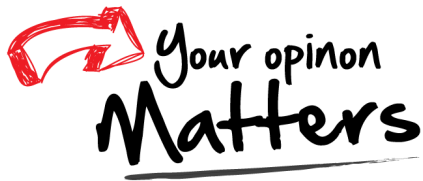
Let  $G$  be a finite group,  $a \in G$ .

Then the order of  $a$  divides the order of  $G$ .

## Theorem

- ▶ There are only 2 groups of order 4.
- ▶ There are only 2 groups of order 6.

Thank  
You!



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